DSA 8420 Spring 2025
Homework 10

Due April 8, 2025

1. Consider the following cost matrix for a transportation problem in which the objective is to minimize the total transportation cost.

	Destination			
Source	1	2	3	Supply
1	8	5	4	50
2	6	8	9	20
Demand	10	20	40	

   (a) (5 points) Write down the linear programming formulation for this problem.

   (b) (5 points) Set up the transportation tableau and use the northwest corner rule to find an initial basic feasible solution.

   (c) (5 points) Write down the dual problem of the linear program in part (a).

   (d) (5 points) Write down all of the complementary slackness conditions.

   (e) (10 points) Solve the problem using the transportation algorithm.

2. Consider the following cost matrix for a transportation problem in which the objective is to minimize the total transportation cost.

	Destination				
Source	1	2	3	4	Supply
1	6	9	4	7	20
2	3	2	5	6	40
3	4	8	2	3	15
Demand	10	25	15	25	

(a) (10 points) Set up the transportation tableau and find an initial basic feasible solution.

(b) (10 points) Beginning with the initial basic feasible solution in part (a), solve the problem using the transportation algorithm.